Alberta
100 MW AI Campus

Strathcona County · 100 MW IT · phased gas-backed development

Develop four 25 MW liquid-cooled halls on a marketed Heartland tract, pairing 100 MW IT with 160 MW net modular gas generation and a conditional transmission connection. Sell committed powered/cooled capacity; earn the right to build each next phase through land, service, permits and tenant evidence.

CONCEPT DESIGN BASIS	LOCAL-CURRENCY FINANCIAL CASE
100 MW IT / four 25 MW phases	CAD 2,268m development CAPEX
Sandhills Heartland candidate - SE 23-56-21-W4	CAD 2,382m funding incl. carry/collateral
Peak site demand 125 MW	Equity IRR: no unique result / zero terminal sale value

Investment stance

Base economics fail the hurdle; advance only bounded diligence, no investment decision

Evidence date: 13 October 2026. Land rights, power capacity, permits, suppliers, customers and financing are not secured. This package is concept/pre-feasibility work, not certified design or construction authorization.

Package contents

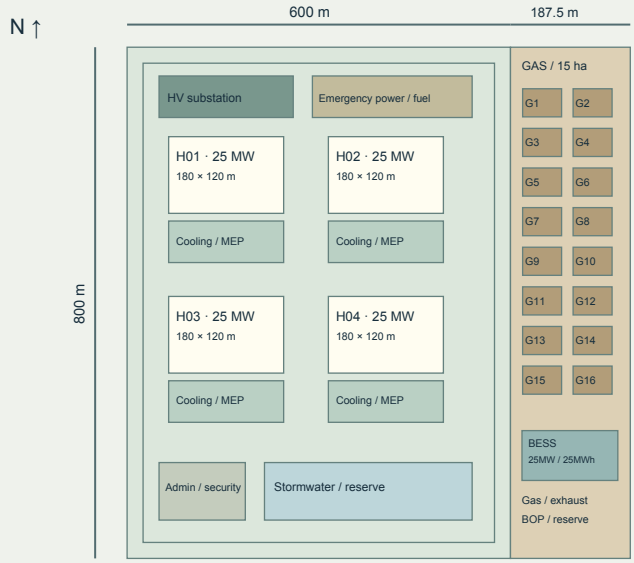
Independent site and power case; concept engineering; permitting and delivery; commercial strategy; procurement; operations and commissioning; evidence and assumptions; calculated cash flows and sensitivities.

01 / Campus test-fit

Editable vector original included in the project package. Geometry, equipment and ratings are explicit concept assumptions.

ALBERTA, CANADA / CONCEPT LAND ALLOCATION

Campus 600 × 800 m + generation 187.5 × 800 m = 63 ha gross



HALL FOOTPRINT

4 × 21,600 m² = 86,400 m²

BUILDING COVERAGE

18% of 48 ha campus envelope

LAYOUT STATUS

Test-fit, not a cadastral overlay

LAND CONTROL

No option, survey or access right

SITE VALIDATION

Title / easements / water / flood / geotech

POWER COMPOUND

16 gas modules + BESS / BOP

DESIGN GATE

Only 1.75 ha gross land remains

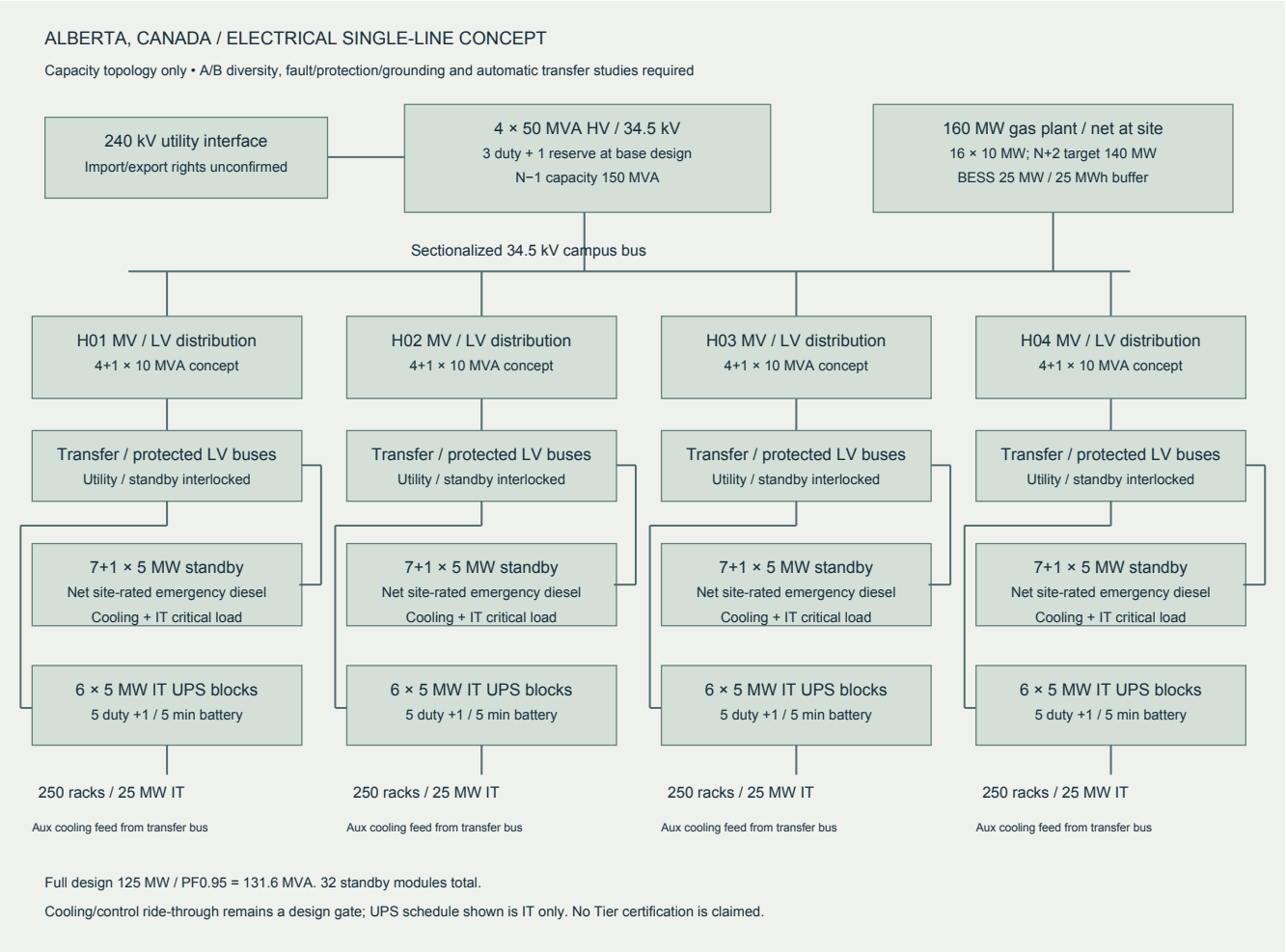
63 ha concept is a tight gross fit on 64.75 ha. Survey, setbacks, wetlands or hazard buffers may disqualify this parcel.

Internal roads, fire access and stormwater must be reconciled to civil engineering. Rectangles are conceptual allocations.

A rectangular land budget tests the approximate program footprint. It is not a surveyed parcel overlay; site-specific easements, environmental setbacks, drainage and road geometry require a revised test-fit.

02 / Electrical single-line

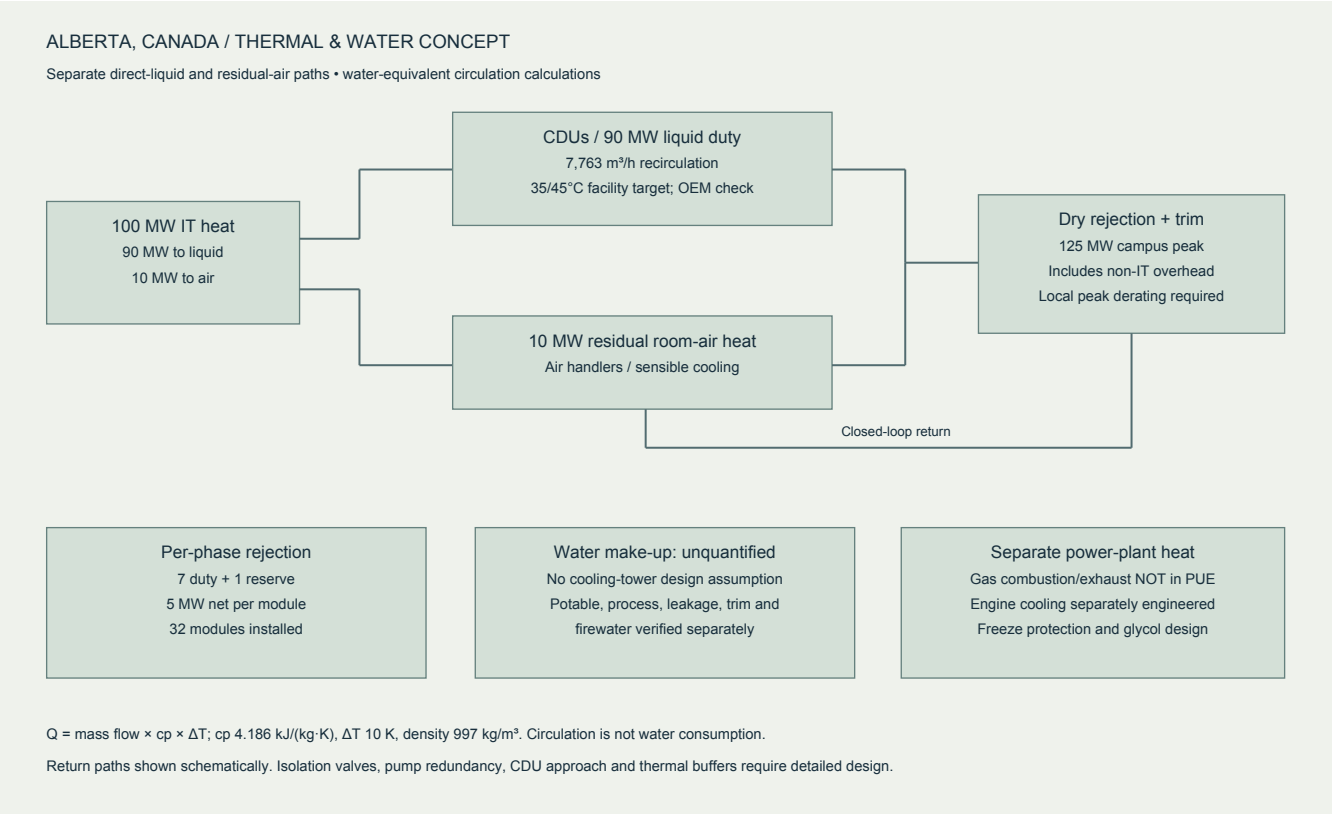
Editable vector original included in the project package. Geometry, equipment and ratings are explicit concept assumptions.



The scheme shows capacity relationships. Isolation, diverse distribution, grounding, fault levels, protection coordination, standby transfer and load acceptance require detailed engineering.

03 / Thermal concept

Editable vector original included in the project package. Geometry, equipment and ratings are explicit concept assumptions.



Annual energy efficiency and peak heat-rejection sizing are separate. Circulating flow is not water consumption. Water availability, local weather design points and manufacturer performance require confirmation.

Heartland AI Campus - Alberta development proposal

Concept / pre-feasibility edition · 13 October 2026 · all monetary amounts CAD unless stated

Decision: the base case fails the 15% return hurdle. Advance only a bounded program to validate site, power, achievable rent and cost; do not take an investment decision. The proposal is a complete development concept for a 100 MW IT campus in Strathcona County, delivered in four 25 MW phases. It is not an application, engineered design, valuation, offer to lease, investment solicitation or representation that the identified property is available.

The asset business is a landlord/operator supplying powered, cooled and connected IT capacity. The tenant buys and operates GPUs and other IT equipment. The model includes facility and proposed generation capital; it excludes GPU/cloud revenue and does not use that revenue to make the property economics work. No landowner, regulator, utility, vendor, customer or lender has been contacted for this study.

Investment proposition and present weakness

A large industrial-area site with new modular generation can support a technically credible Alberta development. The commercial proposition is four separately commissioned buildings, a shared but sectionalized utility backbone, liquid cooling and dry heat rejection, a clearly documented power operating envelope, and expansion rights that turn into construction commitments only when the tenant and financing support them.

The distinction between an attractive Alberta narrative and an executable project is substantial. The initial AESO 1,200 MW large-load allocation is fully assigned; this project holds no allocation. AESO's latest dated BYOG update inspected, September 8, 2026, says final materials were still being completed and intake was expected in fall. That evidence supports developing a BYOG application; it does not support promising access, a qualifying generator ratio, bridging load or a service date. AB02AB03

The chosen site is a real marketed tract, but its listing identifies agricultural zoning. Nearby transmission, a gas franchise and an estimated fibre flag are leads to investigate, not deliverable services. The strongest first product for a prospective partner is therefore a bounded development package with explicit failure tests, not a claim of shovel-ready capacity. AB01

1. Site selection and control strategy

Working candidate: the marketed 160-acre tract at Township Road 564 and Range Road 211, Strathcona County; legal description **SE 23-56-21-W4**. The provincial dashboard places it in the Sandhills Heavy Industrial Policy Area and gives a listing map pin at **53.866399, -113.001632**. The full listing HTML contains that coordinate; it is not a survey coordinate or proof of property boundaries. **160 acres converts to approximately 64.75 hectares.** AB01

Open the candidate location

The same public record reports a nearby AltaLink 240 kV corridor. Geographic proximity warrants a connection study, but does not settle thermal capacity, system stability, access rights, line tapping, route feasibility or cost. The listing's fibre indicator is estimated. Current asking price, seller willingness, title, availability and occupation were not established. All engineering diagrams are conceptual arrangements, never a surveyed overlay. AB01

Land program

Seek a conditional, assignable option over the whole tract, with access for investigations, staged option payments, extension rights tied to public approvals, and exit rights for power, gas, environmental, geotechnical, zoning and customer failures. Obtain title/caveats, legal and beneficial owner confirmation, mortgages, surface/mineral rights, utility and pipeline easements, access rights and property-tax history. Do not purchase speculatively on the strength of the website listing.

Use a **600 m x 800 m campus envelope, 48 ha**, plus an adjoining **187.5 m x 800 m generation zone, 15 ha**: a **787.5 m x 800 m combined envelope, 63 ha gross** within the listed 64.75 ha tract. Only approximately 1.75 ha remains unallocated, so the site fit is tight and must be tested against the actual boundaries, easements and hazards. The campus envelope

includes conceptual drainage and buffers; none is a finding of usable land. Indicative footprint allocation is:

Component	Planning allocation	Purpose
Four hall shells	8.64 ha	Four nominal 180 m x 120 m buildings, 25 MW IT each
Facility electrical and cooling yards	12 ha	Modular services, acoustic separation, maintenance access
Gas plant, engine cooling, exhaust and BESS	15 ha outside campus envelope	Separate plant safety and operating zone
Grid receiving station	3 ha	Concept four-transformer station and service corridors
Roads, loading, security and parking	7 ha	Fire circulation, replacement-equipment routes and construction access
Campus drainage, buffers and reserved space	17.36 ha	Preliminary balance within the 48 ha campus; must expand if field constraints require it

No setbacks in this allocation are statutory claims. A local planner, surveyor, civil engineer and fire authority must determine the permitted arrangement. Buildings should remain separate from the plant acoustic boundary and industrial-neighbour hazard contours. Reserve two independent fibre entries and prevent their construction paths from converging at a single culvert or bridge.

The AG listing and the County's distinction between AG, IHH, IMH and utility zones require written use classification and a mapped zoning check. A heavy-industrial *policy* area does not substitute for the applicable zone or development permit. AB06

Alternatives screened

Candidate	Evidence and practical fit	Decision
55021 Range Road 225, Sturgeon County	First-party broker brochure describes 73.23 acres with I4 industrial zoning, but only about 20 acres remaining for lease. Existing gas/power is not a quantified large-load offer. AB13	Reject for this co-located ultimate campus unless additional land is secured. It may fit a smaller, externally powered block.
Edmonton Energy and Technology Park	The City's current application discussion identifies the chemical-cluster zone as permitting major data-centre and utility uses, while a medium-industrial amendment remains under consideration. No controllable parcel was identified for this proposal. AB14	Retain as a relocation search area. Evaluate an existing suitable zone and actual tract; do not underwrite approval of the pending amendment.
Sandhills candidate	Identified tract large enough to test an integrated layout, but AG zoning and unknown services.	Preferred investigation target, not a confirmed best-value site. Re-select if entitlement, hazards or power fail.

2. Power: selected route and rejected shortcuts

The base concept is **new generation paired with a conditional transmission connection**, with an island-capable design pursued only if studies, approvals, fuel security and customer terms support it. Alberta policy directs data centres toward new supply and cost responsibility for upgrades; that means generation and connection costs must be visible in the project. AB05

Operating quantity	Concept value	Interpretation
Ultimate IT nameplate	100 MW	Four blocks of 25 MW; rentable IT boundary
Annual PUE target	1.20	Energy ratio at stated operating assumptions, not a peak guarantee
Peak facility envelope	125 MW	1.25 x IT; sizing allowance pending equipment guarantees
Proposed generation	160 MW net at site	16 procurement modules targeting 10 MW net each
With two modules unavailable	140 MW	Arithmetic headroom above 125 MW; not a reliability certification
Plant-side storage	25 MW / 25 MWh	Initial dynamic-support allowance, subject to simulation
Grid receiving concept	240/34.5 kV; 4 x 50 MVA	Three duty plus one spare; station within facility electrical cost; connection not selected

The gas modules are installed in cumulative stages of **6 / 9 / 13 / 16** for **25 / 50 / 75 / 100 MW IT**. Physical additions remain **6 / 3 / 4 / 3 engines**, but generation **cost** shares are **47.0486% / 15.8854% / 21.1806% / 15.8854%**. The CAD25m gas lateral and CAD30m BESS are fully funded in phase 1; the other CAD305m follows incremental module counts. This gives phase plant hard costs of **CAD169.375m / CAD57.1875m / CAD76.25m / CAD57.1875m**. Remaining balance-of-plant and civils are allocated proportionally as an approximation; actual deposits and cancellation liabilities are not modeled. At the first stage, four available modules after two outages give 40 MW against a 31.25 MW facility envelope. At later stages that comparison is 70 versus 62.5 MW, 110 versus 93.75 MW, and 140 versus 125 MW. These balances assume **net guaranteed site capacity**, not catalogue rating, and do not prove transient stability or fuel availability.

Manufacturer data demonstrates an appropriate engine size class; published reference performance must be derated for the actual site, auxiliaries, gas and emissions requirements. The independent concept assumes **7.6 GJ/MWh net heat rate**. It does not select a manufacturer or reserve a factory slot. AB11

Alternatives to the selected power route

1. **Grid-only campus:** do not use as the base case without a new allocation and binding service terms. A line nearby does not resolve the large-load process. 2. **Remote generator partnership / eligible paired supply:** retain as a commercial alternative if it offers a cheaper bankable delivery path. It requires the same discipline on eligibility, transmission, risk allocation and net cost. If a partner owns the plant, remove plant CAPEX only while adding the full power/capacity-payment obligation; never remove both. 3. **Fully off-grid operation:** retain as a separately approved fallback, not a shortcut around power-plant, environmental or municipal approvals. Its gas-supply common-mode failure, island dynamics, black-start and tenant SLA requirements are more demanding. 4. **Renewables-plus-storage:** evaluate as emissions and energy-price diversification, without assuming a firm 125 MW round-the-clock obligation is satisfied by annual energy matching. No renewable or export revenues support the base case.

Grid and controls package

Commission the load and generator study together: staged maximum demand, step and cyclic loads, fault ride-through, harmonics, reactive requirements, grounding, protection discrimination, loss of mains, controlled islanding and reconnection. Identify the exact applicable AESO and reliability obligations through the connection process. The AESO technical guide discusses possible 10 MW/min point-of-connection ramp limits and coordination requirements; this is a design prompt, not an approved site-specific operating limit. AB04

A plant BESS may smooth ramps and support an island, but **25 MW cannot absorb or supply a full 125 MW instantaneous imbalance**. The controller must coordinate engine reserve, UPS, load modulation and protective shedding. A 25 MWh energy rating alone does not establish ride-through. The integrated test plan must demonstrate the actual largest credible disturbance and all allowed transitions.

Four transformers arranged N+1 do not prove independent utility supplies. Require the transmission owner to disclose upstream common elements and permitted import/export under outages. Similarly, N+2 engines sharing one gas main do not create diverse fuel. Independent facility emergency supply and full-path UPS remain in the facility scope until a customer-approved reliability analysis allows a different configuration.

Gas and emissions

The working peak gas reservation is **0.8 million standard m3/day**. It is derived from 160 MW x 7.6 GJ/MWh x 24 hours divided by an assumed 38 MJ/m3 gas energy content, rounded upward. At 125 MW actual plant output the corresponding quantity is approximately 0.60 million m3/day. Require a firm delivery and pressure study, maximum-day rights, actual gas assay, curtailment history, upstream outage map, metering/regulation redundancy and lateral easements. A local gas franchise boundary is not a high-pressure industrial service contract.

At 95% committed occupancy and 80% load on committed IT, the campus averages 76 MW IT. At PUE 1.20 it consumes about **799 GWh/year** at the facility electrical boundary. If that entire supply were generated by the assumed engines, a screening factor of 0.38 tCO2/MWh gives roughly **304,000 tonnes of direct CO2 annually**, excluding upstream emissions and methane slip. This is a scenario calculation, not an inventory, guaranteed dispatch pattern or low-carbon claim.

TIER treatment must be developed as a benchmark-based compliance case. The provincial guidance generally covers emitters above 100,000 tonnes CO2e/year and describes benchmark-based obligations. Do not multiply gross emissions by a headline tax and call it the statutory liability. AB10 The May 2026 Canada-Alberta agreement conditionally holds the federal Clean Electricity Regulations in abeyance during litigation; it does not establish a permanent exemption for this plant. Finance counsel must support the operating and emissions path over the financing term. AB16

3. Cooling, water and engineering interfaces

The target product is dense liquid-cooled capacity, with a starting equivalent of **1,000 racks at 100 kW/rack**. This is a capacity equivalence, not a confirmed server layout. Confirm the tenant's liquid fraction, inlet/outlet temperatures, water chemistry, connector type, pressure, allowable transients and responsibility at the CDU. Do not order cooling hardware around an unspecified GPU generation.

Use separate technology-cooling and facility loops, redundant CDUs/pumps and outdoor freeze-protected glycol circuits. Plan a 10 K facility-loop temperature difference for initial hydraulic sizing, subject to customer requirements. As a dimensional check, 100 MW transferred to water at 10 K requires approximately **2.39 m3/s** using 4.18 kJ/kg-K and 1,000 kg/m3; 25 MW is about 0.60 m3/s. Glycol reduces heat capacity and changes pump duty, so this is not a pipe-sizing result. Dual loops, redundancy and design margin increase installed equipment capacity.

The model assigns **90 MW to liquid capture and 10 MW to residual IT air cooling**, with a conservative **125 MW aggregate campus heat-rejection envelope** including peak facility overhead. At the stated 4.186 kJ/kg-K and 997 kg/m3 water assumptions, the 90 MW / 10 K liquid circuit circulates approximately 7,763 m3/hour. Final equipment sizing must locate compressor/fan/pump heat and electrical losses within the actual loops; the 125 MW aggregate is not one mandatory water loop. The gas plant has **separate engine jacket-water, charge-air and exhaust heat systems**; 160 MW electrical generation is not a 160 MW thermal load, and the campus dry-cooler figure cannot cover the engine plant.

Use dry heat rejection with mechanical trim for high ambient temperatures, not evaporative cooling towers in the base design. **+35 deg C summer design check, +40 deg C stress check and -40 deg C winter check are explicit project assumptions**, not sourced design statistics. Obtain hourly dry-bulb/wet-bulb data and appropriate code design values before guaranteeing PUE or capacity. The ECCC climate-normal service is the official starting point, but no site-hourly simulation was performed here. AB12

Winterization must include glycol concentration, separation of indoor water from exposed circuits, idle-equipment freeze risk, restart controls, snow and ice access, heated critical enclosures, drainback and pipe movement. Smoke and dust require filter strategy and differential-pressure monitoring. Redundant mechanical trains must retain maintainability when one train is unavailable at the design envelope.

Zero routine evaporative cooling is an engineering intent, not zero water use. Plan domestic/service supply initially at 50 m3/day, with first fill, maintenance disposal and fire demand separately assessed. Size fire storage and replenishment against the local fire analysis; do not invent a municipal hydrant capacity. Wetlands, drainage changes, construction water and wastewater can require authorization even with dry cooling. AB09

Other deliverables before detailed design are a frost/geotechnical foundation design, stormwater retention and release plan, flood and surface-drainage review, hazardous-neighbour screening, fire vehicle/access design, battery/UPS fire strategy, fuel-spill containment, security zoning and two-carrier route diversity evidence.

4. Permitting and approvals sequence

This proposal separates municipal permission for the campus from permission for its power facilities. AUC Rule 007 provides current facility application requirements and forms for the relevant generation and electrical infrastructure; AUC guidance also identifies separate environmental and noise obligations. AB07AB08 The exact regulatory boundaries and applicable current law must be confirmed for the final configuration.

Workstream	Indicative sequence	Approval evidence required before release
Land and title	M1-6	Conditional control, survey, easements, investigation rights
County classification / rezoning / development	M2-18	Written use interpretation, applicable zoning/development consents and conditions
AESO / transmission-owner process	M2-36 assumption	Final-route eligibility, completed studies, agreements, contributions and operating envelope
AUC generation and electrical facilities	M5-24	Relevant approvals following required application and engagement process
Air/environment and noise	M4-24	Applicable EPEA approvals, dispersion/noise outcomes and operating conditions
Water/wetlands/drainage	M3-20	Activity/diversion approvals where needed, supply and discharge arrangements
Indigenous / participant involvement	Begin M2; through approval	Applicable consultation determination, process completion and documented responses
Gas lateral and roads	M3-36	Supplier and regulator approvals, easements and transport/access permits
Building and safety codes	M14-42, then by phase	Building/trade/pressure/fire compliance and occupancy evidence
Tax and operating compliance	Resolve before finance close	Levy payer/rate treatment, TIER/other reporting and operating register

These month ranges are scheduling assumptions, not regulator service standards. Hearings, consultation, environmental field seasons, route objections and incomplete applications can extend them. Do not underwrite an expedited-approval entitlement or a waived assessment. Begin engagement with nearby land users and potentially affected Indigenous communities early; neither a private-land option nor an industrial policy designation proves consent.

5. Delivery and procurement plan

Month 1 is the authorized start of actual development, not today's date. The base scenario accepts 25 / 50 / 75 / 100 MW at **M42 / M51 / M60 / M69**. It is not an AESO energization forecast. The full month-by-month workstream schedule is in `monthly-schedule.csv`.

The critical path is the latest of (a) land-use/environment/power consents, (b) firm fuel and grid service, (c) manufacture and transport of engines/transformers/switchgear, and (d) construction plus witnessed integrated testing. Hall construction by itself cannot pull that date forward.

Period	Deliverable and decision
M1-3	Site/control screening, title and services map, first fatal-flaw review
M2-8	Provider feasibility, tenant design brief and scoped technical applications
M3-6	Conditional land option and G0 decision; reject the tract if fundamental constraints fail
M6-12	Anchor terms, power/fuel route and budget validation; explicit stop/continue decision
M8-38	Conditional long-lead reservations, engineering and manufacture, with funded cancellation exposure
M12-24	Commercial/FID work, consents and detailed design; conditions must actually be met before irreversible releases
M20-34	Shared civil works and first shell, released only with required approvals
M30-42	Initial 60 MW gas stage and first 25 MW IT fit-out; protection, thermal and operational tests
M36-51	Block 2 and generation expansion to 90 MW cumulative
M45-60	Block 3 and generation expansion to 130 MW cumulative
M54-69	Block 4 and generation expansion to 160 MW cumulative

The aggressive M42 first-phase scenario **depends on equity-funded conditional equipment reservations / limited notices starting around M8**. A 30-month long-lead package then arrives around M38. Those commitments can precede final permits and must carry explicitly approved cancellation exposure; they are not authorized purchases in this artifact. If orders wait until later approvals, move service dates later and rerun the model. Do not preserve the attractive schedule while removing the early-order risk.

Tender equipment and EPC scope through packages with clear interfaces: receiving station; gas lateral; modular engines/BOP; plant BESS/controls; civil/shell; facility electrical/UPS/emergency supply; cooling/CDUs; carriers; integrated controls; and independent commissioning. Planning lead allowances are 30 months for major station equipment and engines, 24 months for gas infrastructure, 18-20 months for major facility systems, and 15 months for diverse fibre work. None is a vendor commitment.

Every major package requires a bid comparison with exclusions, currency, incoterms/delivery point, tax, guarantee conditions, spare parts, escalation, delay damages, warranty, termination rights and service support. Separate factory tests from installation acceptance and integrated performance tests. Require the OEM's net output/heat-rate/noise/emissions guarantees at agreed site conditions; catalogue gross performance cannot satisfy the acceptance schedule.

The independent commissioning authority should own a failure matrix including utility loss, transformer/MV section loss, engine and gas-system trips, BESS unavailability, cooling train/pump loss, communications loss, smoke filtration, UPS bypass and controlled restoration. Customer service starts only after witnessed tests, permits, operating procedures, trained staff and formal acceptance. Concurrent maintenance is a design target, not a Tier certification or a promise of fault

tolerance.

6. Commercial model and customer qualification

The anchor product is a dedicated 25 MW building divided into five 5 MW suites, with paid expansion options over later buildings. Target a 10-15 year take-or-pay capacity lease supported by a parent guarantee or letter of credit. The tenant's credit and delivery obligations matter more than an impressive company name in a prospect list; no prospect interest is assumed. The 20-year operating model assumes continued occupancy after the proposed 10-15 year initial lease expires. A 15-year initial lease leaves five modeled operating years uncontracted for phase 1, and approximately 2.75 for phase 4 under this horizon. No renewal downtime, inducement, rent haircut or tenant-capital replacement is modeled. That tail requires a renewal/reletting downside and cannot be presented to lenders as signed revenue.

The contract must separate:

- Committed IT kW, minimum rent and reservation fees from actual metered IT energy.

- Capacity rent from pass-through energy operating costs and the operating-risk sharing mechanism.

- Landlord facility availability from tenant software/GPU availability.

- Normal maintenance and regulator-directed curtailment from avoidable landlord failure.

- Generation capital recovery from gas, service, carbon and balancing costs.

- Landlord property obligations from the computing-equipment levy and other tenant-specific taxes.

- Capacity acceptance tests from cloud benchmark tests outside the landlord's scope.

Alberta's tax overview states that a levy on large data-centre computing equipment became effective January 1, 2026, with rates up to 2% and lower rates associated with own-power solutions. **GPU exclusion from this model does not establish an exemption.** Tax counsel must identify the legally liable entity, reporting boundary, actual rate and any credit treatment; the lease needs an enforceable reimbursement mechanism where appropriate. No levy saving or tax credit is capitalized in the base facility value. AB15

The go-to-market package is a design brief, constraint map, power-service case, programme, delivered-cost schedule and draft interface/acceptance requirements. Qualify Canadian data-residency requirements, emissions policy, permitted workloads, peak step behavior, density, temperatures, fibre latency and willingness to accept the actual AESO operating envelope. A tenant that requires an incompatible emissions or uptime product is not a fit even at an attractive headline rent.

7. Native-CAD underwriting and cost boundary

All numbers below are **independent screening estimates**, not current vendor quotes, comparable transactions or lender terms. They are designed to expose the cost/rent thresholds that the next diligence phase must validate. Regional assumptions were constructed independently of the Texas proposal.

Input	Base assumption	Estimation basis
Facility hard cost	CAD13,500/kW IT	CAD2,430 shell/fit-out + 5,400 electrical/UPS/standby + 2,970 cooling/CDUs + 1,350 civil/drainage + 675 controls/security/fibre + 675 testing/commissioning
Facility hard cost at 100 MW	CAD1,350m	Excludes land, receiving-grid scope, gas generation, soft cost and contingency
Land	CAD8m	CAD50,000/acre placeholder for whole tract; seller price not established

Input	Base assumption	Estimation basis
Off-plot grid interface	CAD75m	External line/route, reinforcement contributions, easements and utility studies; receiving station already in facility electrical; replace with actual study
Generation + plant BESS	CAD360m	Engine packages 175m + generation BOP 85m + gas lateral 25m + BESS 30m + dedicated plant civils/electrical 45m
Direct subtotal	CAD1,793m	Before soft costs, contingency and model-specific financing treatment
Soft cost	10% of direct	Development, engineering, permitting, legal and owner management
Contingency	15% of hard plus soft cost	Concept reserve; not a probabilistic confidence claim
Fixed non-energy OPEX	CAD35m/year ultimate	Facility/plant staffing 8m, property tax 18m, insurance 4m, fixed plant service 2m, security/admin 3m
Variable non-energy OPEX	CAD180/committed kW-year	Facility maintenance and consumables; no duplicate gas service cost
Capacity rent	CAD235/committed kW-month	Price to test against customer bids; no market quotation represented
Stabilized commitment / actual load	95% / 80%	Distinct rent-bearing capacity and utilization assumptions
Senior debt / rate / amortization	55% / 7.5% / 20 years	Conditional target, not a financing offer
Equity hurdle	15%	Developer screen

The operating energy allowance is **CAD120/MWh** at the facility boundary: delivered fuel CAD41.8 (7.6 GJ/MWh x CAD5.50/GJ), plant variable service CAD17, emissions reserve CAD25, residual grid/transport/balancing CAD21.2, and operations reserve CAD15. These buckets are scenario allowances and must be replaced by an actual dispatch/tariff/fuel model. The emissions reserve is not a calculated statutory TIER bill. Generation capital is already included above and recovered through rent, so a capital-recovery adder must not also be included here.

At the stabilized assumptions, capacity rent is **CAD267.9m/year** (95,000 kW x CAD235 x 12). Fixed plus variable non-energy costs total **CAD52.1m/year**, before any model-specific additional reserves or deductions. Metered facility energy is approximately **798,912 MWh/year**, costing and reimbursing approximately **CAD95.87m/year** at the assumed rate, with no energy trading margin. These arithmetic checks are not debt service, net profit or an IRR. The interactive model provides the time-dependent investment outputs and its explicit financing/terminal assumptions.

The model applies soft cost to direct cost, then contingency to direct plus soft: CAD1,793m + CAD179.3m + CAD295.845m = **CAD2,268.145m before financing**. Do not silently add an allowance twice or call unfinanced construction cost total equity funding. Debt needs depend on draw timing, capitalized interest/fees, debt-service reserves, and operating carry during phase-up. Tax, depreciation, GST financing/recovery and entity-specific levy incidence require a separate tax/cash model before finance close.

Current model result and investment decision

The executable base case yields **negative CAD419.26m equity NPV at a 15% hurdle**, with **CAD2,382.25m total modeled funding including CAD114.11m of capitalized interest**. Stabilized EBITDA is CAD215.8m and cash available for debt service after the model maintenance-capital reserve is CAD198.7m. These values are generated from the same assumptions used by the interactive model; the conclusion is **no investment decision at current price and cost assumptions**.

The equity IRR is **undefined as a single investment measure**: the model detects multiple roots, so the displayed IRR must remain unavailable rather than selecting the flattering root. The unlevered IRR is about **5.20%**. Minimum ordinary operating DSCR is approximately **1.27 x** , but minimum DSCR **including the terminal repayment is only 0.45 x** . A **CAD130.72m debt balloon** remains when the model ends because later-phase 20-year loans extend beyond a horizon measured 20 operating years from the first phase. That is a terminal funding requirement, not ordinary operating default evidence by itself.

The model assigns **zero terminal enterprise/land value**, assumes constant nominal inputs and excludes tax benefits; these choices matter to valuation. A future sale or refinance cannot be treated as committed funding. The capacity rent required to reach zero equity NPV at the chosen 15% hurdle is approximately **CAD339.28/kW-month**, versus the assumed CAD235. That is a calculated threshold to test, **not an assertion that customers will pay it**. An executable case needs demonstrably better cost, price, timing or risk structure, followed by a fresh investment decision.

The fixed OPEX estimate explicitly includes plant staffing and fixed service. The CAD18m property-tax allowance is informed by published 2026 Strathcona rates: approximately CAD900m assumed non-residential assessment and CAD400m assumed machinery/equipment assessment produce roughly CAD18m tax before small additional adjustments. Those assessment values are assumptions, not cost-to-assessment equivalence or a County ruling. No industrial tax incentive is assumed. AB17

Finance and downside discipline

Use a sponsor holding company with facility and generation SPVs, documented easements and a power-service agreement. Consolidate both asset costs for investment evaluation. A partner's separate capital is still an economic cost if recovered in power or capacity payments.

Debt should follow a signed credit-supported anchor lease, service certainty, required permits, firm EPC scope and completion support. The sponsor must fund early diligence and any equipment cancellation risk before lenders are willing to fund. Later blocks require their own tenant and capacity gates; one tenant's non-binding expansion intention cannot secure debt for the full campus.

At minimum, test total cost +35%, capacity rent -20%, occupancy 70%, first service +12/+24 months, delivered energy CAD180/MWh, lost pass-through recovery, interest +200 basis points, and plant replacement/overhaul. Energy pass-through protects the landlord only if the customer can afford and is contractually required to pay it. A combined downside can invalidate a seemingly healthy stabilized cash yield. Do not use terminal value, merchant exports, incentives, heat sales or speculative AI revenues to rescue a weak contracted case.

8. Risk register, gates and next decisions

Decisive risk	Evidence needed	Stop or reshape trigger
Land/entitlement	Option, title, survey and County interpretation	No controllable/consentable tract by M6
BYOG / grid	Current eligibility, viable studies, costs and operating terms	No written viable route by M12 or credible first service beyond M54
Gas	Firm peak-day delivery and pressure / outage study	Required fuel cannot be secured on bankable terms
Receptors / environment	Noise, air, wetland and industrial hazard findings	Material conflict cannot be mitigated within layout and cost
Anchor credit / rent	Signed 25 MW take-or-pay and acceptable security	No bankable anchor by M12 at viable economics
Supply chain	OEM slots, net guarantees and bounded cancellation exposure	Manufacturing dates or exclusions defeat the delivery case
Common-mode outages	Fuel/grid dependency and integrated failure-mode analysis	Customer availability terms cannot be achieved
Carbon / levy	Current legal and tax opinion; enforceable customer allocation	Finance-term operating or customer emissions requirements cannot be met
Fibre	Physically distinct carrier routes and measured service design	Single shared failure point cannot be addressed economically
Financial downside	Updated integrated model against bids and leases	15% equity screen or lender downside coverage not achieved

G0 requires controlled land and a developable-area test. **G1** requires a credible power/fuel route. **G2** requires a consentable layout. **G3** requires bankable first-phase customer and financing terms. **G4** is the investment decision with actual conditions met, funded risk and executable scopes. **G5** is witnessed readiness and customer acceptance. Every gate remains open.

The immediate work is to verify the offering and seller terms; prepare a combined AESO/TFO/generation pre-application package; obtain firm-gas and carrier feasibility; seek County written use classification; commission environmental/geotechnical fatal-flaw work; obtain budgetary equipment scopes; and test tenant pricing/credit against the proposed product. These are a prioritized diligence plan, not actions already performed. A sensible first paid work package would be capped and milestone-based, with title/site, power/gas, environment and customer teams working in parallel before detailed design expands.

Evidence and assumptions

`project.json` is the machine-readable source register, assumption ledger, project definition, risks and gates. `monthly-schedule.csv` gives 69 monthly workstream records. The estimates above are separate from public-source facts. Access dates are **2026-10-13**; undated pages are marked undated rather than given invented publication dates. Public marketing pages do not prove availability, and regulator policy pages do not grant this project rights.

AB01 - 160 acres: Township Road 564 and Range Road 211

Government of Alberta Regional Dashboard. Published/effective: Undated live listing. Accessed 2026-10-13.

Supports: Listing identifies SE 23-56-21-W4, 160 acres, AG zoning and Sandhills Heavy Industrial Policy Area; reports a 240 kV AltaLink line 1.35 km away and estimated fibre availability. HTML map marker is 53.8663990960219,

-113.001631865825.

Limit: Listing is evidence of a marketed candidate, not current availability, ownership, a survey, utility capacity or connection rights. Distances are straight-line estimates.

AB02 - Large Load Projects

AESO. Published/effective: Live page; guide posted 2026-06-12. Accessed 2026-10-13.

Supports: Phase 1 has allocated all 1,200 MW: GLDC 970 MW and Keephills Phase I 230 MW; other applications enter Phase 2.

Limit: No capacity allocated to this proposal; named projects are unrelated third parties.

AB03 - Phase 2A Large Load Integration: BYOG Process

AESO. Published/effective: 2026-09-08 latest dated stakeholder update inspected. Accessed 2026-10-13.

Supports: September update says final BYOG process and materials are still being finalized and intake is expected in fall; earlier draft outcomes are explicitly not final.

Limit: This is the published page state inspected, not confirmation intake remains closed on every other AESO channel. Final ratios, qualifying generation, tethering and milestones need confirmation.

AB04 - Guide to AESO Connection Requirements for Transmission-Connected Data Centres

AESO. Published/effective: 2026-06-12. Accessed 2026-10-13.

Supports: Guide covers load dynamics, protection, backup synchronization, operational coordination and possible 10 MW/min point-of-connection ramp limits; its examples are illustrative.

Limit: Project requirements emerge from studies and binding agreements. This proposal has no validated dynamic model or approved ramp envelope.

AB05 - Powering new pathways for data centres

Government of Alberta. Published/effective: Royal assent 2025-12-11. Accessed 2026-10-13.

Supports: Bill 8 framework encourages own generation and cost-causation for transmission upgrades.

Limit: Policy direction does not establish approved project service, final tariff or an energization date.

AB06 - Strathcona land use bylaw: zones and maps

Strathcona County. Published/effective: Live consolidated page. Accessed 2026-10-13.

Supports: County distinguishes AG, Heavy Industrial Heartland (IHH), Medium Industrial Heartland (IMH), utilities and overlays; mapped zoning governs.

Limit: Does not determine a data-centre/generation use classification or rezone the candidate. Seek written interpretation and current parcel zoning certificate.

AB07 - Rule 007: Facility Applications

Alberta Utilities Commission. Published/effective: Effective 2025-11-06. Accessed 2026-10-13.

Supports: Current AUC facility application rule and forms cover thermal generation, transmission and substations.

Limit: Timing and hearing requirements are case-specific; none of this proposal has been filed.

AB08 - Proposing a power plant application

Alberta Utilities Commission. Published/effective: Live guidance. Accessed 2026-10-13.

Supports: AUC guidance addresses facility applications, affected persons, Rule 012 noise and separate EPEA requirements for thermal plants above 1 MW.

Limit: General guidance; exact industrial, generation, pipeline and environmental approval boundaries require regulator and counsel confirmation.

AB09 - Water legislation and guidelines

Government of Alberta. Published/effective: Live guidance. Accessed 2026-10-13.

Supports: Water Act/EPEA framework includes diversion licences, activity approvals, wetlands, wastewater and stormwater guidance.

Limit: Dry cooling does not exempt site drainage, wetland disturbance, construction water or domestic supply from applicable requirements.

AB10 - Technology Innovation and Emissions Reduction Regulation

Government of Alberta. Published/effective: Includes December 2025 amendments. Accessed 2026-10-13.

Supports: TIER generally covers facilities emitting at least 100,000 tonnes CO2e per year; compliance is benchmark-based, not simply gross emissions times a headline tax.

Limit: Plant-specific treatment, benchmark, credits and compliance prices are unresolved; no assumed exemption or incentive.

AB11 - Wartsila 31SG gas engine product technical data

Wartsila Energy. Published/effective: Live product page. Accessed 2026-10-13.

Supports: Manufacturer publishes 60 Hz units around 10.4-12.4 MW and heat rates around 6.9-7.1 GJ/MWh at specified reference conditions.

Limit: A technology reference, not a selected supplier, quote, delivery slot or site guarantee; auxiliaries, climate, emissions and fuel change net output.

AB12 - Canadian Climate Normals 1991-2020

Environment and Climate Change Canada. Published/effective: 1991-2020 dataset. Accessed 2026-10-13.

Supports: Official normals use composite stations with at least 15 years of observations; appropriate starting point for an hourly site weather study.

Limit: No hourly simulation or site design extremes extracted here. All design temperatures and PUE values in this proposal are assumptions, not values attributed to this page.

AB13 - Sturgeon County Industrial Land: 55021 Range Road 225

Royal Park Realty. Published/effective: 2025 brochure. Accessed 2026-10-13.

Supports: Broker advertises 73.23-acre industrial site with only about 20 acres remaining for lease and I4 zoning.

Limit: First-party marketing can be stale; service capacity is unstated and no owner contact occurred.

AB14 - LDA26-0225: Edmonton Energy and Technology Park AI Data Centre Amendment

City of Edmonton. Published/effective: 2026 live application consultation. Accessed 2026-10-13.

Supports: City describes large AI data centres as allowed in the chemical-cluster zone but currently not allowed in the medium-industrial zone; an amendment is under consideration.

Limit: An application is not an approval. No specific available EETP parcel is secured or represented by this proposal.

AB15 - Alberta tax overview: data centre levy

Government of Alberta, Treasury Board and Finance. Published/effective: Effective 2026-01-01 per live page. Accessed 2026-10-13.

Supports: Provincial overview says equipment levy effective January 1, 2026, up to 2%, with lower rates for projects supplying their own power and corporate-tax treatment.

Limit: Facility SPV versus tenant liability, rate formula and tax-credit realization need tax counsel. Exclusion of GPU CAPEX from this model is not a levy exemption.

AB16 - Canada-Alberta Implementation Agreement

Prime Minister of Canada. Published/effective: 2026-05-15. Accessed 2026-10-13.

Supports: Agreement conditionally holds Clean Electricity Regulations in abeyance while the reference and possible appeals proceed; contemplates subsequent equivalency negotiations if upheld.

Limit: Does not guarantee permanent exemption or a 20-year plant operating entitlement. Read with current legislation, litigation and project-specific advice.

AB17 - 2026 Strathcona County tax rates

Strathcona County. Published/effective: Approved 2026-04-28. Accessed 2026-10-13.

Supports: 2026 non-residential rate 15.0823 mills and machinery/equipment 11.0960 mills, with designated industrial requisition listed separately.

Limit: Assessment class and taxable values for this unbuilt project are unknown; no exemption or tax incentive assumed.

Alberta execution and acceptance package

13 October 2026 · concept issue · all gates open. This execution basis accompanies the Alberta dossier, electrical/cooling/layout drawings and executable financial model. It defines what evidence would support a construction and service decision; it does not claim vendor compliance, a booked production slot, an approved operating mode, a lease or an energized site. Project months are conditional planning intervals and carry no calendar-date promise.

The selected design envelope is four 25 MW IT phases, **125 MW campus peak** at an assumed 1.25 peak multiplier, 90 MW ultimate direct-liquid IT heat and 10 MW residual-air IT heat. Ultimate primary generation is **16 x 10 MW net at site**, with cumulative 6 / 9 / 13 / 16 modules for the four phases. With two modules unavailable, those stages provide 40 / 70 / 110 / 140 MW against phase campus peaks 31.25 / 62.5 / 93.75 / 125 MW. This static balance does not prove transient response, gas availability or permissible island operation. Independent emergency diesel remains separate from primary generation.

Physical engine staging and financial cost staging are different. The CAD360m primary-plant allowance places **the full CAD25m gas lateral and CAD30m BESS allowance in phase 1**; the remaining CAD305m is apportioned by 6 / 3 / 4 / 3 additional engines. The gas lateral and complete 25 MW / 25 MWh BESS concept are therefore funded before initial

operation, not progressively introduced with the last engines.

Phase / assumed operation	Added engines / cumulative	Engine-related share of CAD305m	Shared gas lateral + BESS	Allocated plant hard cost
1 / M42	6 / 6	CAD114.375m	CAD55.000m	CAD169.375m
2 / M51	3 / 9	CAD57.1875m	-	CAD57.1875m
3 / M60	4 / 13	CAD76.250m	-	CAD76.250m
4 / M69	3 / 16	CAD57.1875m	-	CAD57.1875m

The executable model uses those phase cost fractions (approximately 47.0486% / 15.8854% / 21.1806% / 15.8854%) and applies soft cost and contingency separately. Remaining engine balance of plant and dedicated civil/electrical work within CAD305m still follow a proportional approximation; common infrastructure may need further front-loading after a scoped quote. Each phase's allocated cost is spread evenly across its modeled construction interval. Actual supplier deposits, reservation payments, milestone releases and cancellation liabilities are not modeled; a phase-level cost allocation does not prove adequate liquidity on every procurement date.

Conditions for issuing and releasing procurement

Issue budgetary RFQs during site/power diligence using the same input revision, operating envelope and drawings. Each response must show exclusions, net site ratings, ambient and altitude derating, emissions/noise basis, guaranteed performance, equipment weights/footprints, interface ownership, lead time from an explicit release event, price validity, escalation, cancellation exposure, spare parts, factory testing and on-site support. No response is presumed received.

The M42 first-operation target requires **conditional long-lead release around M8**, because 30 months of manufacture/delivery then runs to M38, leaving approximately four months for final installation, integration and acceptance. A slot reservation that does not start engineering/manufacture is not equivalent to an effective production release. These early commitments would precede some permits and financing conditions. They therefore need explicit sponsor risk approval, capped loss exposure, cancellation/transfer rights where negotiable and sufficient equity; approval in this artifact does not authorize any purchase.

If releases must wait until M24 permit completion, a 30-month package arrives no earlier than about M54 before installation/commissioning. In that case M42 fails and the entire plan and cash model must be rebaselined, provisionally to at least M58 rather than implying vendor acceleration. The executable model presently uses a simplified monthly spend curve and does not represent actual reservation deposits or cancellation liabilities.

Package	RFQ issue / clarification target	Conditional award or effective release	Site delivery / readiness target	Install and acceptance target	Release evidence
240 kV receiving transformers, protection and interconnection engineering	M2-M5	M8 for 30-month package	M38; foundations/haul route ready earlier	M38-M40 installation; M40-M42 authorized energization and integrated tests	TFO interface accepted; design freeze, access rights, permit/cancellation plan, executable price and manufacture-start obligation

Package	RFQ issue / clarification target	Conditional award or effective release	Site delivery / readiness target	Install and acceptance target	Release evidence
First six gas modules and relevant plant balance of plant	M2-M6	M8 for 30-month package	M38; foundations, gas train and exhaust interfaces ready	M38-M40 erection; M40-M42 plant and campus trials	Net ratings at site; emissions/noise basis; fuel specification and firm service pathway; LTSA and guarantees; acceptable permit exposure
Gas lateral, metering, regulation and firm transport	M3-M6	M12 for 24-month development package	Service-ready target M36, subject to route consents and gas-provider agreement	M36-M40 leak/pressure/control tests and authorized gas-in	Written pressure, capacity, quality and maximum-day duty; easements; upstream interruption terms; no inference from nearby pipe
Facility MV/LV, UPS, batteries and emergency diesel	M6-M10	M16 for 20-month package	M36	M36-M40 installation; M40-M42 integrated tests	Approved discrimination/grounding basis and IT/cooling ride-through scope; emergency-only operating envelope; rating guarantees
BESS, PCS, EMS and plant controller	M6-M10	M18 for 18-month package	M36	M36-M40 installation; M40-M42 dynamic tests	25 MW / 25 MWh remains a placeholder until time-domain study; usable energy, SOC policy, grid-forming need, fire/cyber strategy
Liquid cooling, air cooling and dry rejection / trim	M6-M10	M18 for 18-month package	M36	M36-M40 hydraulic/electrical work; M40-M42 thermal tests	Tenant water-class and residual-air requirements; site-condition capacity curves; freeze protection and mechanical trim duty
Civil works and first hall shell	M8-M12	M18 enabling scope; field work only after approvals, target M20	Shell/weather-tight target M34	M30-M40 phased MEP fit-out overlapping approved areas	Survey, geotechnics, setbacks, drainage, fire access, issued-for-construction drawings and permits
Two carrier routes and entries	M4-M8	M18 for 15-month allowance	M33 service-ready target	M33-M40 independent route tests; M40-M42 tenant acceptance	Carrier route drawings and third-party crossings; confirmed physically separate paths and repair service terms
Independent commissioning agent	M6-M9	M10 design-review appointment	Test procedures and load-bank plan M30-M36	M36-M42 progressively witnessed completion	Independence, written acceptance criteria, approved safe test modes, calibrated instrumentation and customer/TFO witnesses

Dates are target gates, not supplier quotes. Later plant stages add **3 engines by M51, 4 by M60 and 3 by M69**. Applying the same 30-month supply plus four-month field-test allowance requires effective releases around **M17, M26 and M35**, respectively. Later release or scope change shifts the corresponding phase; an unexercised framework option is not evidence of manufacturing capacity. Confirmed customer reservations, cancellation security and financing must cover those early expansion commitments. Later civil/MEP execution can overlap operation only with proven physical segregation and approved management of change.

Commissioning and acceptance matrix

The numeric test durations and excursion envelope below are **proposed contractual acceptance targets**. OEM ratings, tenant tolerance, permits and system-operator requirements may require different tests. Any missing threshold is a hold point for agreeing the protocol, not permission to pass. The independent commissioning authority records procedure revision, prerequisites, calibrated instruments, event/time-series data, defects, retests and signed disposition. Testing that could affect the grid occurs only within authorized switching and operating instructions.

Test / objective	Proposed measurable acceptance evidence	Witness and hold point
Equipment and construction completion	100% of safety-critical punch items closed; approved as-builts, torque/insulation/pressure results, relay settings and cause/effect schedules match installed equipment; no unresolved issue blocks safe isolation	Engineer of record, EPC quality lead and independent commissioning authority; no energization before electrical/gas/fire permissions
Net generation rating	Test each engine's net delivered MW at the agreed ambient, altitude, fuel pressure/composition and auxiliary load; contract target 10 MW net/module. Correct to agreed conditions using vendor-approved curves; demonstrate sustained stable operation and record fuel/heat rate/emissions	OEM and independent test engineer; no assumed guarantee from gross nameplate or brochure; site-condition acceptance method agreed before award
First phase continuous campus duty	25 MW metered IT load-bank demand plus measured campus overhead, up to 31.25 MW planning peak, for proposed 24-hour integrated run after thermal stabilization. No unplanned IT power interruption; all agreed electrical and coolant envelopes maintained	Tenant technical representative and commissioning authority; final test load and duration agreed in lease acceptance schedule
Ultimate campus duty	Repeat cumulative acceptance at 50/75/100 MW IT, with measured supporting power. Ultimate design envelope 125 MW facility input; at least the proposed 24-hour run at final-stage duty, supported by approved heat-rejection operating conditions	Tenant, EPC and independent authority. Actual IT load/thermal simulators needed; low-load energization is not proof of full duty
Primary N+2 static capacity	At each stage demonstrate the full commissioned campus duty with two primary modules unavailable: available design capacity 40/70/110/140 MW. Record every online module's net MW, reserve margin, temperature and fuel conditions	OEM, commissioning authority and operator; this capacity test does not imply tolerance of all common gas/controller faults

Test / objective	Proposed measurable acceptance evidence	Witness and hold point
Dynamic load steps and generator loss	Study and safely test staged ±5 MW IT-equivalent load changes and loss of the largest operating 10 MW generation block as initial planning disturbances; prove bus voltage/frequency and recovery remain inside the **agreed OEM, tenant and operator envelope** with no critical-load dropout	TFO/AESO where applicable, OEM, tenant and commissioning authority; exact ramp, nadir, overshoot and time limits must be signed before acceptance
Grid disturbance and separation	If islanding is expressly approved, test authorized utility loss, controlled separation, island stabilization and synchronized reconnection at agreed loading. Capture relay actions, BESS power/SOC, bus quality and IT continuity; no unauthorized backfeed	Utility/system operator and licensed switching authority; approved studies and protection/control settings are prerequisites. If islanding is disallowed, revise the SLA and operating case
Black start and emergency recovery	In an approved dead-bus procedure, establish controls, auxiliaries, generation and cooling in the correct sequence, then restore tenant load within the contracted restoration envelope. Demonstrate an independent recovery path when a plant controller or one start source is unavailable	OEM, site operations and commissioning authority; 25 MWh BESS capacity alone does not establish black-start sufficiency
IT UPS and cooling/control ride-through	Test one redundant UPS module unavailable while the phase carries 25 MW IT; demonstrate the proposed five-minute IT ride-through using measured delivered battery energy and load. Separately validate control/CDU/pump and residual-air cooling continuity or thermal hold-up during the measured transfer/start interval	Tenant, UPS OEM, cooling OEM and commissioning authority; 8.333 MWh full-campus delivered IT energy is a sizing identity, not an installed battery acceptance certificate
Thermal and hydraulic performance	Prove each 25 MW phase rejects 22.5 MW through the direct-liquid path and 2.5 MW through residual-air cooling at assumed 90/10 capture, while total rejection meets measured overhead-inclusive duty. Verify supply/return temperature, flow, differential pressure, water chemistry, leak alarms and balancing; test one cooling/CDU/pump module unavailable	Cooling OEM and tenant; proposed 35/45 deg C facility loop requires tenant/OEM sign-off. Outdoor design/stress performance requires curves, tests or an agreed seasonal completion plan
Freeze and seasonal operation	Validate outdoor fluid freeze protection against agreed -40 deg C winter screening condition, cold-start and low-load controls, drainage and heat trace; validate +35 deg C design / +40 deg C stress screening against site weather basis and net equipment curves	Mechanical engineer and OEM; assumptions are not local design-weather certification or a guarantee that tests occur at those outdoor temperatures
Fuel service and safe shutdown	Verify metered pressure/quality/flow across the agreed maximum-day envelope; alarm and safely isolate on gas-quality/pressure excursions; prove gas detection, ventilation, fire interfaces and emergency shutoff	Gas provider, plant OEM, fire/safety authority as applicable; physically separate primary modules do not remove shared upstream-fuel risk

Test / objective	Proposed measurable acceptance evidence	Witness and hold point
Emissions, noise and water compliance	Independent measurements/reports satisfy the actual permit limits and operating restrictions; close all monitoring/reporting prerequisites. Record operational gas and water meters, discharge controls and spill-response equipment	Environmental lead and relevant authority; numeric permit limits must come from the issued approvals, not this concept
Facility maintenance and A/B path isolation	Execute agreed maintenance scenarios with safe locked-out equipment while critical loads remain served; trace all shared buses, controls, pipes, fuel and communication paths; document failures outside the maintainability claim	Operator, engineer of record and commissioning authority; module counts alone do not establish a certified topology
Network and controls	Trace two off-site routes to independently located carrier infrastructure; prove agreed throughput/latency/failover and alarm delivery. Demonstrate privileged access, backup/restore, time synchronization and OT segmentation without live unauthorized intrusion	Tenant network lead, carriers and controls owner; logical carrier diversity is insufficient if the physical route is shared
Operational handover	Approved SOP/MOP/EOP set, asset register, warranty starts, spare inventory, on-call matrix and training records; conduct tabletop plus controlled emergency drills; signed tenant acceptance with only agreed noncritical residual items	Site general manager and tenant; commissioning documents and live staffing must exist before ready-for-service declaration

Operating organization and controls

The initial organization is an **indicative staffing plan to cost**, not a claim that the current OPEX allowance has been supported by payroll quotes. A proposed 24/7 core shift comprises a shift lead, a qualified plant/control-room operator, an electrical/critical-facilities technician and a mechanical/cooling technician. Four continuously covered posts require roughly 20-24 trained FTE at a 5-6 FTE relief factor per post, before supervision, dedicated maintenance and security. Professional licensure, power-engineering classification and safe staffing must be confirmed against the selected equipment and applicable Alberta requirements. Do not count an on-call vendor as an on-site shift operator.

Accountable role	Recurring deliverable and escalation
Site general manager	Own service performance, staffing, safety and incident decisions; approve operating readiness and management of change; reconcile budget to the agreed SLA and plant duty
Shift lead / control room	24/7 power, thermal, gas and alarm supervision; signed rounds/logs; dispatch only within the approved import/export and emissions envelope; trigger EOP on threatened limits
Electrical lead	Maintain protection settings, battery health, switchgear/transformer condition, isolation permits and switching authority; plan PM windows against live capacity and redundancy
Mechanical / cooling lead	Track flow, pressure, chemistry, filters, fluid condition and thermal margins; maintain CDUs/pumps/dry coolers/trim; seasonal freeze and heat-stress readiness
Gas-plant lead and LTSA provider	Per-engine run hours/starts, oil analysis, fault trends, OEM bulletins and scheduled overhaul plan; maintain net-rated capacity after planned outages; track exclusions and response commitments
Fuel and energy manager	Firm gas nominations/capacity and pressure risk; meter reconciliation, supply/transport terms, cold-day contingency; ensure pass-through bills do not duplicate rent-funded capital or fixed O&M

Accountable role	Recurring deliverable and escalation
Environmental and safety lead	Permit monitoring, emissions inventory/reporting, noise/water obligations, spill/fire preparedness, contractor controls and regulator notifications under issued conditions
Controls / OT and network owner	Access reviews, offline recovery copies, approved patch/testing windows, control-network segmentation, time-series retention and diverse carrier evidence
Maintenance planner / stores	CMMS asset register and OEM PM calendar, critical-spare min/max stock, calibration, warranty claims, service bulletins and turnaround logistics
Customer service / commercial owner	Capacity and energy billing, SLA/incident reporting, planned-work notices, expansion reservations, credit security and dispute escalation

Security, HSE, procurement, day maintenance and management are additional costed roles or contracted services. The financial model's fixed OPEX line must be reconciled to this organization and the selected LTSA before underwriting approval. A minimum availability promise cannot rely on staffing or replacement stock that has not been contracted.

The plant LTSA RFQ should define planned service/overhaul intervals, consumables, major component swaps, local response times, guaranteed parts access, performance degradation, fuel exclusions, emissions-system upkeep, liquidated damages and boundaries between warranty, capital overhaul and variable maintenance. Model maintenance capital is a smoothing allowance; it does not place physical spares on site or replace an overhaul cash calendar.

Critical-spares screening includes CDU/pump controls and seals, sensors/meters, valve actuators, protection relays, UPS modules and control boards, battery strings or replacement arrangements, engine ignition/fuel components, filters and fluids, and agreed long-lead transformer/engine exchange strategy. Quantities must follow failure consequence, supplier lead time and actual installed models. Track stock and replenishment; do not assume a spare capacity module contains every replaceable part.

Every intrusive task uses an approved method of procedure with affected load, isolation/lockout, remaining capacity, rollback and witnesses. Major events use immediate safety stabilization, tenant/system-operator/regulator notification according to the actual contracts/permits, preserved time-series/event logs, root-cause review and a tested corrective action. The concept proposes monthly reliability/PM/capacity review and pre-winter/pre-summer readiness reviews; mandatory reporting cadence comes from the final agreements and approvals.

Anchor-customer term-sheet scope

This is a negotiation checklist and proposed allocation, not a sent offer or legal advice. Counsel must translate it into enforceable facility, energy, easement and intercreditor documents.

Term	Proposed commercial position / evidence needed
Demise and capacity	Initial 25 MW IT block, optionally divisible into 5 MW suites, at a precise IT delivery demarcation; roof/yard, security, service access and fit-out responsibilities explicitly scheduled
Rent and duration	Underwriting input CAD235 per committed kW-month; proposed 10-15-year take-or-pay term. It is **not** a cleared price or proof the 15% hurdle is met. Reprice against the live model and lender requirements before FID
Metered power	Pass actual agreed variable operating energy cost through separately; define meters, losses/PUE allocation, audit rights, imbalance/demand costs, gas/carbon exposure, taxes and dispute mechanism. No hidden second recovery of owner-financed generation capital
Indexation and costs	Negotiate indexed rent and OPEX treatment; executable base model assumes zero escalation. Explicitly allocate equipment levy, tax changes and carbon costs; no tax-credit realization assumed
Tenant interfaces	Rack loading/density, cooling class/chemistry/flow/pressure, water ownership, residual-air load, voltage/quality, dynamic load changes, firmware/control compatibility and fiber requirements confirmed before design freeze

Term	Proposed commercial position / evidence needed
Ready for service	Tied to signed acceptance protocol and permits, not a utility energization notice alone. Phase-specific long-stop, tenant access, test loads, punch-list handling and early-access liability negotiated
Availability and curtailment	SLA based on approved and tested topology and fuel/grid modes; specify planned maintenance, lawful curtailment, tenant load-step limits and service credits. Do not promise off-grid independence before approved island tests
Security and remedies	Credit-approved parent guarantee, cash security or LC; capped delay/service remedies, termination and cure rights; lender direct agreement and step-in rights negotiated without assuming bankability
Expansion	Separate, time-limited and funded options for later 25 MW phases; reservation deadlines aligned with engine manufacture releases and tenant cancellation exposure; avoid building speculative phases on nonbinding interest
Data and sustainability	Customer data-residency/security obligations distinct from facility marketing; disclose gas generation, emissions calculation boundary and renewable claims only to the extent contractually evidenced
Acceptance of economics	Current base-case returns and terminal debt obligations supplied with the term sheet; if lease price or risk allocation fails underwriting, reopen economics rather than describe a conditional customer as committed

Current open evidence

Before a financed notice to proceed, obtain the actual site survey and usable acreage, approved BYOG/grid operating envelope, gas service agreement, applicable planning/environmental/plant approvals, signed OEM performance/lead-time/LTSA terms, constructible coordinated design, tenant technical schedule and credit-backed lease, independently reconciled budget/contingency, staffing and overhaul plan, and lender-approved funding/covenants. All remain open. The proposal is complete as a development concept only; external readiness requires these documents and witnessed results.

Calculated financial appendix

All amounts below are CAD millions except rent, ratios and capacity. Generated from the same executable model used by the website. Pre-tax, un-escalated planning inputs; no terminal sale proceeds.

Development capital	CAD million
Building shell and fit-out	243.00
Campus electrical, UPS and emergency diesel	540.00
Cooling, heat rejection and CDUs	297.00
On-plot civil works and drainage	135.00
Controls, security and campus fiber	67.50
Installation testing and commissioning	67.50
Land acquisition allowance	8.00
Off-plot grid connection and utility works	75.00
Dedicated primary generation plant	360.00

Development capital	CAD million
Design, development, legal and permitting	179.30
Concept contingency	295.85
Total development CAPEX	2268.14
Capitalized construction interest	114.11
Refundable utility collateral	0.00
Total funding requirement	2382.25
Construction equity	1020.67

Stabilized lease revenue: 267.90m/year. EBITDA: 215.80m/year. Customer rent plus metered energy: 319.10 per committed kW/month. Equity IRR: No unique IRR (multiple-roots). Equity NPV at 15%: -419.26m. Minimum annual operating DSCR: 1.27x. Minimum annual DSCR including terminal debt balloon: 0.45x. Terminal debt balloon: 130.72m. Rent for zero equity NPV: 339.28 per kW/month.

Annual cash flows

Year	End IT MW	CAPEX	EBITDA	Debt service	Equity cash flow
1	0	104.99	0.00	0.00	-47.25
2	0	187.02	0.00	0.00	-84.16
3	0	550.44	0.00	0.00	-247.70
4	25	728.13	31.47	22.46	-321.61
5	75	531.14	103.40	66.56	-210.87
6	100	166.43	179.83	111.17	-20.69
7	100	0.00	215.80	131.63	67.07
8	100	0.00	215.80	131.63	67.07
9	100	0.00	215.80	131.63	67.07
10	100	0.00	215.80	131.63	67.07
11	100	0.00	215.80	131.63	67.07
12	100	0.00	215.80	131.63	67.07
13	100	0.00	215.80	131.63	67.07
14	100	0.00	215.80	131.63	67.07
15	100	0.00	215.80	131.63	67.07
16	100	0.00	215.80	131.63	67.07
17	100	0.00	215.80	131.63	67.07
18	100	0.00	215.80	131.63	67.07
19	100	0.00	215.80	131.63	67.07
20	100	0.00	215.80	131.63	67.07
21	100	0.00	215.80	131.63	67.07
22	100	0.00	215.80	131.63	67.07
23	100	0.00	215.80	131.63	67.07
24	100	0.00	89.92	185.56	-102.77

Sensitivities

Independent one-variable changes. Energy cost affects customer cost and passes through landlord revenue/expense at cost.

Variable	Low / high input	Low / high equity NPV
Base rent	188.00 / 282.00	-608.2 / -230.3
Facility construction cost	10800.00 / 16200.00	-245.8 / -592.7
Committed occupancy	80.00 / 100.00	-558.9 / -372.7
Pass-through electricity cost	84.00 / 156.00	-419.3 / -419.3
Construction and commissioning shift	30.00 / 54.00	-473.4 / -372.1
Loan interest	5.50 / 9.50	-340.7 / -506.8

Model qualifications

- Concept estimates, not contractor quotes, grid rights, permit approvals or a financing offer.

- Pre-tax, constant nominal inputs: no inflation/escalation, corporate/income tax, depreciation, grants, terminal asset value or GPU purchases.

- PUE is an annual input, not an hourly weather simulation; low-load fixed power and demand ratchets are not modeled.

- Electricity is passed through at cost; changes affect tenant cost, not landlord EBITDA. Owned-plant energy price must contain variable generation costs only.

- No lease ramp: each commissioned phase immediately reaches the chosen occupancy. Schedule is conditional, not a forecast of approval.

- Returns use monthly timing; annual DSCR uses cash available after maintenance capital, including any final debt balloon.

- Debt extends beyond the project horizon: the final cash flow includes repayment of the remaining principal.

- Primary generation plant is separately owned in this scenario. Its fixed O&M must be included in fixed OPEX; major overhauls and fuel/carbon exposure require dedicated diligence.

- Pre-tax, un-escalated planning model; no depreciation, income tax, incentives or GPU assets.

- No terminal sale proceeds or land salvage; utility collateral returns at the modeled horizon.

- No lease-up ramp: occupancy applies from each phase commissioning date.

- Maintenance reserve is preliminary; plant major overhauls and tenant equipment levy require separate diligence.

Full input register

Input	Value
energyPriceMwh	120
basePue	1.2
peakPue	1.25
facilityCapexPerKw	13500
landCostM	8
gridCostM	75
powerPlantCostM	360
softCostPct	10
contingencyPct	15
fixedOpexM	35
variableOpexPerKwYear	180
leaseRateKwMonth	235
debtPct	55
interestPct	7.5
debtYears	20
equityDiscountPct	15
occupancyPct	95
loadFactorPct	80
firstOperationMonth	42
phaseIntervalMonths	9
constructionMonths	24
operatingYears	20
maintenanceCapexPct	1
utilityCollateralM	0
generationPhaseShares	0.4704861111111111,0.15885416666666666,0.21180555555555555,0.15885416666666666
rackKw	100
liquidCapturePct	90

Input	Value
coolingDeltaTC	10
coolantSpecificHeatKjKgK	4.186
coolantDensityKgM3	997
powerFactor	0.95
electricalLossPct	4
batteryMinutes	5

Engineering and financial methodology

Common model and engineering basis

Status: concept / pre-feasibility, 13 October 2026. This is an inspectable calculation basis for two separate development proposals. Every equipment selection, cost, financing term and schedule here is a design assumption unless a project dossier gives a supporting source. It is not a stamped design, power reservation, cost quote, Tier certification, construction authorization or underwriting opinion. Each project retains its own native currency, inputs and commercial/power risks.

Scope and accounting boundary

Each campus has **100 MW of IT capacity in four 25 MW phases**. The modeled business owns and leases powered/cooled facility capacity. It does not buy GPUs or sell cloud compute. Tenant occupancy is contracted IT kW; tenant load factor is average metered IT consumption divided by contracted IT capacity. Those quantities are independent.

The facility hard-cost input includes the building, on-plot civil works, transformers, UPS, an **independent emergency diesel system**, cooling, controls, campus fiber and commissioning. Land and off-plot utility works are separate. If `powerPlantCostM > 0`, that is a separate continuously operating primary plant, also owned by the project. It does not replace the emergency diesel equipment in this design. Removing diesel from an alternative proposal requires changing the equipment scope and facility hard-cost estimate, not just changing the narrative. Both normal plant and standby plant require separate emissions/noise/fire reviews.

For a grid-fed project, `energyPriceMwh` is a blended delivered energy allowance, including any tariff components allocated per consumed MWh. For an owner-operated generation project, it means **variable fuel, variable maintenance, emissions/carbon and any grid energy expense per delivered facility MWh**. It must not include the primary plant's capital recovery or fixed O&M: capital is already in `powerPlantCostM`, and fixed plant O&M must be in `fixedOpexM`. That boundary avoids charging for the plant twice. Uncontracted power-price risk, peak tariff ratchets, take-or-pay gas and pass-through enforceability remain material commercial questions.

Calculation and API contract

`model.mjs` is browser-compatible JavaScript ESM, with no dependencies. All monetary outputs ending in `M` are **millions of the project's native currency**. Return percentages are percentage points, not fractions. Capacity is MW IT unless specified otherwise. All exports merge `defaults`, `project.modelOverrides`, then explicit scenario `overrides`; inputs are validated without silent type coercion.

import { defaults, calculate, annualCashFlows, sensitivity, engineering } from './model.mjs'; const result = calculate(project, { occupancyPct: 85 });

Export / output	Meaning
`defaults`	Frozen numerical fallback assumptions; regional overrides are the actual project basis.
`calculate(project, overrides)`	`{currency, inputs, capex, stabilized, finance, timeline, cashFlows, warnings}`.
`capex`	`facilityM, landM, gridM, powerPlantM, hardCostM, softCostM, contingencyM, totalM, breakdown[]`. Breakdown rows contain `id, label, amountM, basis`. Excludes IDC and refundable collateral.
`stabilized`	`occupiedItMw, averageItMw, averageFacilityMw, annualItMwh, annualFacilityMwh, baseRentM, electricityRevenueM, energyCostM, totalRevenueM, fixedOpexM, variableOpexM, ebitdaM, maintenanceCapexM, cfadsM, customerAllInKwMonth`.
`finance`	IDC, funding, debt draw, construction equity, peak cumulative equity deficit, total equity contributions, equity/unlevered NPV and IRR, IRR status, minimum annual DSCR, equity/unlevered break-even rent, yield on construction cost, terminal value/debt balloon, collateral and total cash flows. See exact names in the source.
`timeline`	`firstOperationMonth, finalOperationMonth, lastModelMonth, operatingYears, phases[]`. Each phase has phase number, IT MW, cumulative IT MW, construction start and operation month.
`annualCashFlows(project, overrides)`	Same annual array as `calculate().cashFlows`, without computing IRRs.
`cashFlows[]`	Model year, first/last month in year, operating months, annual-equivalent average online IT MW, ending online IT MW, facility MWh, ending debt balance and DSCR. Cash columns: `capexM, collateralDepositM, collateralReturnM, debtDrawM, capitalizedInterestM, baseRentM, electricityRevenueM, energyCostM, fixedOpexM, variableOpexM, ebitdaM, maintenanceCapexM, cfadsM, interestM, principalM, balloonM, debtServiceM, unleveredCashFlowM, equityCashFlowM`.
`sensitivity(project, overrides)`	Six one-variable-at-a-time rows with `variable, label, unit, low, base, high, lowNpvM, baseNpvM, highNpvM, lowIrrPct, highIrrPct, lowCustomerCostKwMonth, highCustomerCostKwMonth`. Other inputs remain at the scenario basis.
`engineering(project, overrides)`	Design MW/MVA, PUE, rack count, electrical losses, plus `transformer, phaseTransformer, ups, standby, cooling, qualifications`. Engineering sizes to `peakPue`; operating energy uses `basePue`.

Undefined financial ratios are `null`, never fabricated as zero or Infinity. In particular no-debt DSCR is null; zero-occupancy break-even rent and tenant cost are null; IRR without a unique root found in the scan domain is null. Error messages identify the offending input. The API does not assert that valid numeric inputs are commercially or technically achievable.

Operating equations

At full commissioned campus size:

Occupied IT MW = 100 x occupancy / 100 Average IT MW = occupied IT MW x load factor / 100 Annual IT MWh = average IT MW x 8,760 Annual facility MWh = annual IT MWh x annual PUE Base rent = occupied IT MW x 1,000 x rent per kW-month x 12 Electricity revenue = energy expense = annual facility MWh x price per MWh Variable operating expense = occupied IT kW x variable expense per kW-year EBITDA = base rent + electricity revenue - electricity expense - fixed OPEX - variable OPEX Maintenance capital = (facility + primary-generation hard cost) x annual maintenance-capital rate CFADS = EBITDA - maintenance capital

Each commissioned phase immediately reaches the chosen steady occupancy; no customer ramp, bad debt, outage, SLA credit or collection lag is assumed. Fixed OPEX and maintenance capital scale with commissioned phases, including at zero occupancy. Actual sitewide staffing and insurance may be more front-loaded. At zero tenant load the model has zero metered energy: actual unloaded UPS, security, lighting and mechanical energy would worsen the result. Annual PUE is constant through occupancy/load scenarios and requires low-load and hourly validation. All years use 8,760 hours; no leap-year effect is modeled.

Pass-through changes deliberately do **not** create landlord EBITDA. Energy cost still changes the customer's all-in payment and competitive viability. `customerAllInKwMonth` divides base rent plus energy by committed kW-month, not metered kWh.

The maintenance reserve is a level annual capital allowance, not an OEM overhaul forecast or debt-funded expense. Fixed OPEX must cover operating labor, insurance, property/local taxes, security, routine facilities/plant maintenance and administration; variable OPEX must not duplicate the variable costs inside the energy input.

Construction, financing and valuation

Hard cost equals facility cost plus land, grid connection and primary generation. Soft cost is a percentage of **all hard costs including land**; contingency is a percentage of **hard plus soft costs**. This mechanical base is explicitly visible to avoid ambiguous percentage application. Soft cost is not a loan fee. Facility hard cost is allocated 18% shell, 40% electrical/UPS/standby, 22% cooling, 10% civil, 5% controls/security/fiber and 5% commissioning; these are concept allocations, not a quantity-survey estimate.

The engine runs monthly, beginning at M0. Land and off-plot utility cost, including their proportional soft cost and contingency, are spent uniformly M1-M12. Facility cost is allocated equally to four phases. Primary-generation cost uses `generationPhaseShares`, four validated **cost fractions**, summing to one; equal quarters are the fallback. These are distinct from the fraction of physical gas modules installed in each phase.

Alberta installs 6 / 9 / 13 / 16 cumulative gas modules, or incremental engine-count fractions 6/16, 3/16, 4/16 and 3/16. Its CAD360m plant allowance separately allocates **all CAD55m of shared gas lateral (CAD25m) and BESS (CAD30m) to phase 1**, because both must be available for first operation. The remaining CAD305m is allocated by incremental engine counts, giving phase plant costs **CAD169.375m / 57.1875m / 76.25m / 57.1875m**. Accordingly `generationPhaseShares` is `[0.4704861111111111, 0.15885416666666666, 0.21180555555555555, 0.15885416666666666]`. These cost shares do not alter the 6 / 9 / 13 / 16 module schedule. Engine-related balance of plant and dedicated civil/electrical works within the CAD305m still follow a proportional approximation; a quantity-based procurement estimate may require additional early common expenditure.

Facility and allocated plant costs are each spent evenly during the relevant phase's `constructionMonths`, ending one month before that phase's operation month. The full gas lateral and BESS allowance is therefore spent before first operation, but its exact supplier deposit, manufacturing milestone and site-payment timing is **not modeled**. Maintenance capital follows commissioned facility and commissioned plant cost separately. All land/grid common costs are allocated equally to the four phase loan tranches. This assumes early, at-risk common expenditure and does **not** implement the real project's conditional capital-release gates; actual land options, reservation deposits, cancellation exposure, civil sequencing and grid security requirements need a procurement-specific funding curve. The schedule sensitivity moves both phased construction spending and operation dates, while common M1-M12 costs stay fixed. It is not a sunk-cost delay scenario: deferring a negative-NPV project may make its present loss smaller, which does not establish a benefit from construction delay.

Each tranche draws `debtPct` of its construction spending; equity funds the remainder. Monthly interest is `interestPct / 12` times opening balance plus half of that month's new draw. Before a phase operates, interest is capitalized and fully debt financed. IDC is shown separately from construction CAPEX, added to financed principal, and is not also subtracted as an equity-funded cost. Therefore `debtPct` is the proportion of construction CAPEX financed, **not** the final debt-to-total-funding ratio. A lender has not committed to financing IDC.

Starting in the phase's first operation month, its loan pays level monthly principal and interest over `debtYears`. At zero interest, payments are straight-line principal. There is no interest-only operating period, refinancing or sculpting. A remaining balance at the model endpoint is explicitly repaid as a terminal balloon. Monthly identities are preserved in annual aggregation:

Closing debt = opening debt + construction draws + capitalized interest - principal - final balloon Unlevered CF = CFADS - construction CAPEX - collateral deposit + collateral refund Equity CF = unlevered CF + construction debt draws - cash interest - principal - final balloon

Utility collateral is assumed cash funded by equity at M1, locked throughout the model, and refunded at the endpoint; it is not a cost, rent or depreciable asset. This is a conservative liquidity convention, not a contractual statement about when a utility returns security. Earlier milestone release, letter-of-credit alternatives, interest on cash and financing fees are excluded. `constructionEquityM` excludes subsequent operating losses; `maximumCumulativeEquityDeficitM` shows the largest cumulative net equity funding requirement assuming interim cash is retained, while `totalEquityContributionsM` totals negative equity cash flows without netting interim distributions. Neither includes additional lender reserves or discretionary working capital.

The horizon is **20 operating years from the first phase**, plus construction. Later phases therefore have fewer than 20 modeled operating years. Partial calendar years are retained. This deliberately assigns **zero terminal enterprise value and zero land residual value**, with no terminal sale costs or decommissioning expense. Debt is still fully settled. Expanding the horizon changes that assumption; a resale multiple is not hidden in returns.

NPV discounts each monthly cash flow using `(1 + annual hurdle)^(month/12)`. Both equity and unlevered NPV use the displayed `equityDiscountPct` as a common user-selected hurdle. The unlevered number is not an independently estimated WACC valuation. IRR is an annual effective rate calculated from monthly timing. The solver scans annual log-growth rates from -99.99% to 10,000%, bisects sign-changing roots, and reports a value only when one scanned root is found. Nonconventional cash flows can have multiple roots; the scan is not a symbolic proof against arbitrarily close/tangent roots, and `equityIrrStatus` / `unleveredIrrStatus` expose that qualification.

DSCR is annual CFADS divided by that year's cash interest plus principal, including a terminal balloon. Construction equity is not in the numerator. The minimum is across years with debt service; cash-flow insolvency is not hidden by clipping negative ratios. `minOperatingDscr` separately excludes the terminal balloon from the denominator, so normal operating coverage and terminal liquidity can be distinguished. Neither is a lender covenant definition. Yield on construction cost is stabilized EBITDA / CAPEX, before maintenance capital and IDC. Break-even base rent solves a **zero equity NPV at the chosen hurdle**; the separate unlevered break-even solves zero unlevered NPV. It is not merely an operating break-even.

Cash flows are nominal local-currency amounts with **zero escalation/inflation assumed**. Corporate/income tax, tax depreciation, sales-tax timing, depreciation shields, tax incentives/grants, loan/arrangement fees, debt-service reserve, construction revenue, receivable/payable working capital, compute hardware and revenue, terminal asset value and decommissioning costs are excluded. The outputs are pre-tax scenario results. No FX comparison is performed.

Electrical and thermal concept

Item	Calculation / sizing rule
IT	4 x 25 MW; planning density 100 kW per rack yields 250 rack positions per phase and 1,000 total. Other rack densities round positions upward per phase while capacity remains 25 MW.
Peak facility input	`100 MW x peakPue`. Annual average PUE is used only for energy. Texas input 1.40, for example, implies 140 MW at design peak, independent of contracted occupancy. This is a preliminary sizing proxy, not an equipment/weather load calculation.
Electrical overhead	Default 4 MW of IT-relative electrical losses, with the remaining peak overhead assigned to cooling/auxiliaries. Losses are included within PUE; not added a second time.
Primary substation	50 MVA units; duty count is ceiling of peak MW / power factor / 50; add one spare. A 140 MW, 0.95 PF example requires 147.37 MVA and 3 duty + 1 spare, 200 MVA installed / 150 MVA after one loss. Power factor 0.95 is an assumption.
Phase transformation	10 MVA units; duty count is ceiling of phase peak MVA / 10; add one spare per phase. A 35 MW phase at 0.95 PF has 36.84 MVA duty: 4 duty + 1 spare, 20 installed units campus-wide.
IT UPS	5 x 5 MW duty + 1 x 5 MW spare per phase. 120 MW installed, 100 MW firm IT. Five minutes at 100 MW is 8.333 MWh delivered; at assumed 90% delivery efficiency, 9.259 MWh stored. End-of-life, discharge-rate, temperature and reserve margins must be separately specified.
Standby	Net-at-site 5 MW emergency-diesel blocks; ceiling of phase peak MW / 5 plus one spare per phase. At 140 MW campus peak this is 7 duty + 1 spare per phase: 32 units / 160 MW installed. Nameplate catalog capacity cannot substitute for derated net site capacity. Fuel runtime and refueling are unquantified.
IT liquid capture	Assumed 90% of IT heat enters liquid, leaving 10 MW residual air heat at full IT load. Rack vendors must confirm that split.
Liquid loop	Preliminary facility supply/return design **35/45 deg C**, 10 K rise. 35 deg C is within the maximum supply associated with ASHRAE W40, subject to selected IT/OEM acceptance. CDU secondary temperature and approach must be verified; thermal class alone is not an equipment warranty.
Indoor liquid flow	`mass flow kg/s = liquid kW / (4.186 kJ/kg-K x 10 K)`. At 90 MW: 2,150.024 kg/s. Using assumed density 997 kg/m3 gives 7,763.376 m3/h. This is circulating clean water, not consumption or utility withdrawal.
CDU modules	2.5 MW units; at 22.5 MW liquid duty per phase, 9 duty + 1 spare; 40 units total. Installed CDU rating must be at actual liquid temperatures and flow.
Heat rejection	Conservatively size the campus rejection boundary to all peak facility power. At 140 MW, a clean-water-equivalent 10 K circuit is 3,344.481 kg/s or 12,076.363 m3/h. Some losses are actually dissipated outdoors; this is an aggregate upper-bound heat balance, not a single mandatory loop flow.
Heat-rejection blocks	Net-at-design-ambient 5 MW blocks; ceiling of phase heat duty / 5 plus one spare per phase. At 140 MW: 7 duty + 1 spare per phase, 160 MW installed. Regional dry-cooler/mechanical-trim design must deliver that rating at local peak conditions.

Outdoor freeze-protected glycol has different heat capacity, density and viscosity from the clean-water reference. The `coolantSpecificHeatKjKgK` and `coolantDensityKgM3` inputs allow alternate preliminary checks; actual fluid concentration, minimum outdoor temperature, pressure drop, pipe diameters, pump curves and heat-exchanger approach require engineering. A 10 K temperature rise is not evidence that free cooling works in summer. Residual room-air cooling needs a separate appropriately rated air-handling/chilled-water path; the 35 deg C liquid supply is not a room-air cooling set point. Texas's mechanical trim and Alberta's freeze protection have different requirements even though their conservation equations are the same.

The full 100 MW of IT electricity ultimately becomes heat; facility electrical and mechanical consumption adds heat within the boundary. **Primary gas-plant exhaust/radiator heat and fuel energy are outside data-center PUE** and require

separate plant heat rejection/emissions design. Dry cooling does not justify a zero total-water claim: potable use, leakage, flushing, fire reserve, and any optional adiabatic cooling remain unresolved. Annual WUE cannot be inferred from loop circulation.

N+1 quantities prove only installed capacity after one module outage. They do not prove concurrent maintainability, fault tolerance or certification. The engineering package still needs sectionalized buses/valves, maintenance isolation, diverse A/B distribution, rack transfer compatibility, selectivity, grounding, arc-flash/fault studies, harmonic/dynamic-load studies, generator start and load acceptance, cooling/control ride-through and thermal buffer sizing. The UPS schedule currently reserves IT ride-through; it does not silently claim the cooling system is UPS-backed. The commercial availability SLA is conditional on those studies and a commissioned failure-mode test plan.

Evidence and limits

Primary materials accessed 13 October 2026:

1. US Department of Energy, Cooling Water Efficiency Opportunities for Federal Data Centers. Supports annual PUE = facility energy / IT energy, WUE = water / IT energy, the distinction between circulating systems and evaporative consumption, and separation of direct-liquid and residual-air paths. It does **not** support this project's numerical PUE, cost, schedule or water allowance. 2. ASHRAE Handbook, Chapter 20: Data Centers and Telecommunication Facilities. Supports rack-level load characterization, liquid-cooling interfaces, dependence on workload/low-load behavior, and the need to coordinate reliability across facility systems. Its water-class discussion identifies class labels by maximum facility supply temperature. It does **not** certify the proposed rack density, redundancy or OEM compatibility. 3. ASHRAE, 2021 Thermal Guidelines liquid-cooling reference card. Identifies W40 among facility water-supply classes. The proposed 35/45 deg C operation remains an assumption requiring equipment confirmation.

Costs and financial terms are project estimates, not drawn from those thermal references. The documentation separates source-supported definitions from author-created sizing and arithmetic so the proposal can be reviewed without confusing an assumption with a vendor commitment.

Verification

Run `node --test tests/model.test.mjs`. Tests reconcile CAPEX and monthly capacity exposure, source/use cash flows, IDC/debt balance and amortization, electricity pass-through, zero-debt and zero-interest cases, zero occupancy and invalid inputs, negative and undefined IRRs, zero-NPV break-even rents, refundable collateral and terminal balloon, and independent electrical/hydraulic/thermal equations. Sensitivities check the direction of economically relevant changes. A passing test establishes calculation consistency, not external feasibility.